GWCD Documentation - Overview

Vermont AHS GIS and Data Viz | Groundwater Chemistry Database (GWCD)

Groundwater Chemistry Database (GWCD) — Documentation

This repo documents the data source, ETL pipeline, and public-facing product for Vermont's **private drinking water** groundwater chemistry data. It exists so that other teams (e.g. grant-reporting / outreach dashboards) can build on the same data without re-deriving filtering and exceedance logic from scratch.

Live product: Drinking Water Quality (ArcGIS Experience Builder app, public)

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02-etl-pipeline.md	Step-by-step walkthrough of the weekly ETL script
03-filtering-criteria.md	Record inclusion/exclusion rules, with pseudo-code
04-exceedance-calculations.md	How exceedance rates are calculated, threshold table, with pseudo-code
05-geocoding.md	How records get a town assignment; geocoded vs. non-geocodable data
06-application.md	What the public app actually displays, confirmed against its live config
scripts/GroundwaterChemistryDatabasePROD.sanitized.py	Sanitized copy of the production ETL script (credentials/internal hostnames redacted)

TL;DR

- **Source**: `groundwater.ChemistryResults` (SQL Server), loaded weekly into `GroundwaterChemistry_GWCDrawdata` (full, unfiltered, ungeocoded).
- **Scope**: private drinking water wells/springs only (public/municipal water sources are excluded), Vermont addresses only.
- **Output tables**:

    - `GroundwaterChemistry_GWCDPoints` — all records that pass the quality/scope filters, geocoded (address or town-centroid fallback), with exceedance flags and per-town/analyte summary stats.
    - `GroundwaterChemistry_GWCDPointTableExternal` — the above, further deduplicated per location and with town/analyte combinations under 20 samples suppressed. This is what feeds the public app.
- **Public app** currently surfaces 6 analytes: Arsenic, Fluoride, Gross Alpha, Nitrate, Radon, Uranium.
- Script owner of record: originally Victoria Treto, rewritten 2025-03-24 by Daniel Jarvis. Runs weekly via `GroundwaterChemistryDatabasePROD.py`.

Who to contact

AHS GIS & Data Viz team: ADS.VCGIAHSTechs@vermont.gov